

Auditing AI-Based ESG Scores: How Much Can You Trust a Machine-Generated Sustainability Score?

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Abstract

Environmental, Social, and Governance ratings are now central to sustainable investment, yet the AI systems that generate them operate without routine audit. Providers routinely disagree on scores for the same firm, text classifiers used in ESG pipelines can be manipulated through strategic disclosure language, and whole sectors may be structurally disadvantaged without detection. Prior work on ESG scoring has focused on predictive accuracy and rater divergence across providers, leaving internal score consistency, adversarial robustness, auditability of closed models, sector-level fairness, and distribution-shift robustness insufficiently examined within a unified framework. This paper addresses those five gaps by applying empirical stress tests to two public datasets: 503 S&P 500 firm-level ESG risk scores from Sustainalytics and approximately 2,600 labelled sustainability-report sentences from the A3CG greenwashing benchmark. A Spearman rank correlation analysis, an adversarial text-injection experiment, a Random Forest surrogate model, a disparate-impact audit, and a leave-one-sector-out cross-validation are applied across five research questions. We find that a single rater’s four sub-scores rank firms near-independently (average Spearman rho equals 0.12), that greenwashing buzzword injection drops classifier F1 from 0.72 to 0.53, and that an open surrogate reproduces proprietary risk-level labels at 88 percent accuracy. Energy and Utilities firms almost never reach the low-risk band under the 4/5ths disparate-impact rule, while Financial Services and Real Estate produce zero F1 under sector holdout. These findings demonstrate that standard accuracy evaluation systematically misses fragilities that matter for governance, regulatory compliance under the EU AI Act, and responsible capital allocation.

Index Terms

Responsible AI, AI Governance, ESG Auditing, Greenwashing Detection, Algorithmic Fairness, Zero-Knowledge Proofs, Adversarial Robustness, Sustainability, Machine Learning

I. INTRODUCTION

Environmental, Social, and Governance scoring has become one of the most consequential applications of machine learning in finance. Major institutional investors, pension funds, and sovereign wealth funds now use AI-generated ESG scores to construct index weightings, screen bond eligibility, and set lending covenants. The AI systems underlying these scores combine large language models, transformer-based text classifiers, ensemble regressors, and proprietary neural networks to process millions of corporate disclosures, news feeds, and regulatory filings every day. The output of this infrastructure – a single numerical score – is treated by markets as an objective measure of a firm’s sustainability performance. This paper asks whether that trust is warranted.

The importance of this question extends well beyond financial markets. ESG scores are increasingly used by regulators to assess corporate compliance with the EU Sustainable Finance Disclosure Regulation and the EU Taxonomy, by governments to allocate green infrastructure financing, and by civil society organisations to hold corporations publicly accountable for environmental and social commitments. When these scores are unreliable or systematically biased, the consequences fall on communities facing climate risk, workers whose firms are misclassified, and investors whose capital is misdirected. AI systems that influence these outcomes at scale carry governance obligations that standard model evaluation frameworks do not address.

Several structural problems complicate trust in AI-based ESG scoring. First, raters disagree sharply: pairwise Spearman correlations across major providers range from 0.38 to 0.71 [1], suggesting that scores measure different constructs or apply different weightings to the same data. Second, the NLP pipelines used to classify sustainability disclosures are vulnerable to adversarial manipulation, meaning corporations can insert strategic buzzwords to influence their scores without genuine practice change – a form of automated greenwashing that current systems cannot reliably detect [2]. Third, proprietary architectures prevent independent audit under the EU AI Act [3], creating a regulatory gap between what the law requires and what the market currently provides. Fourth, disparate algorithmic treatment of sectors – penalising energy firms regardless of genuine decarbonisation effort – may misallocate capital away from the transition investments that sustainability policy most urgently requires. These concerns belong at the intersection of AI, power, and responsibility: they are not technical edge cases but governance failures with distributional consequences.

Prior work has examined cross-provider ESG rating divergence [1], [4]–[6], greenwashing detection using NLP [7], [8], and fairness in algorithmic decision-making [9]–[11]. However, these streams of work remain disconnected. Studies on rating divergence do not test whether a single rater’s own sub-scores are internally consistent. Studies on NLP robustness do not extend to greenwashing-specific adversarial attacks applied to realistic sustainability-report text. Studies on fairness do not distinguish between sectoral and firm-size unfairness in ESG risk classification. No prior work has demonstrated that a proprietary ESG risk-level function can be surrogated with high fidelity from its own published sub-scores, enabling zero-knowledge proof compliance under the EU AI Act. The absence of a unified multi-dimensional audit framework for AI-based ESG systems is the gap this paper addresses.

This paper implements five empirical audit checks against AI-based ESG scoring and analyses the results through the lens of responsible AI governance. The contributions are as follows:

- This paper measures internal sub-score consistency within a single ESG rater using Spearman rank correlation across 430 S&P 500 firms.
- This paper evaluates adversarial robustness of an ESG text classifier under synonym substitution, character-level noise, and greenwashing injection attacks using the A3CG benchmark.
- This paper demonstrates surrogate-model auditability of a closed ESG risk function and maps the results to zero-knowledge proof compliance pathways under the EU AI Act.
- This paper audits sector-level and firm-size disparate impact in ESG risk classification using the 4/5ths rule.
- This paper tests distribution-shift robustness via leave-one-sector-out cross-validation and identifies sectors where model performance collapses.
- This paper discusses the implications of all five findings for AI governance, power, accountability, and sustainable finance regulation.

II. RELATED WORK AND BACKGROUND

Machine learning methods have been applied to ESG scoring and sustainability assessment across a wide range of architectures. Early work used logistic regression and support vector machines on TF-IDF-vectorised sustainability reports to classify greenwashing claims [2]. Domain-adapted transformer models such as FinBERT [7] and ESG-BERT [8] subsequently improved classification accuracy on financial and sustainability text. Ensemble methods including Random Forest and Gradient Boosting have been applied to structured ESG risk score prediction [12], [13]. Large language models are increasingly used for zero-shot and few-shot ESG claim evaluation [14]. However, the adversarial robustness of these models to greenwashing-specific attacks has not been systematically evaluated prior to the A3CG benchmark [2].

The ESG application domain introduces domain-specific challenges that general NLP evaluation frameworks do not address. Corporate sustainability reports are strategically authored documents designed to maximise favourable perceptions, creating an inherent tension between disclosure and transparency [15]. ESG data availability is unequal across markets and firm sizes [16], meaning models trained primarily on large-cap developed-market firms may perform poorly when applied to smaller or emerging-market firms. Provider-level rating divergence has been documented extensively [1], [4]–[6], but within-provider sub-score consistency has received less attention. The regulatory context has also changed substantially: the EU Sustainable Finance Disclosure Regulation [17] and the EU AI Act [3] now create direct legal obligations for transparency and auditability that ESG AI systems in scope must satisfy.

Responsible AI principles including fairness, accountability, transparency, and explainability [18]–[21] apply with particular force to ESG scoring because the system outputs directly influence capital allocation at scale. Algorithmic fairness research has shown that models can produce structurally unequal outcomes across demographic or categorical groups even without explicit discriminatory features [9]–[11]. Applied to ESG, this raises the question of whether energy or materials sector firms face systematically lower classification probabilities of receiving favourable risk labels, independent of their actual sustainability performance. Zero-knowledge proofs for AI [22], [23] provide a technically grounded path to regulatory compliance under EU AI Act Article 13 – enabling third-party audit without weight disclosure – but require that a sufficiently accurate surrogate model can be constructed from observable outputs. The environmental cost of large AI models is also relevant in this context: deploying computationally expensive transformers for ESG scoring adds to the very carbon footprint that ESG metrics are meant to measure [24], [25].

Table I compares closely related studies across the key dimensions addressed in this paper. The table shows that while individual dimensions such as accuracy, fairness, or robustness have been examined in isolation, no prior study applies all five audit dimensions simultaneously to AI-based ESG scoring systems using reproducible public datasets.

The table shows that prior work has addressed individual audit dimensions but none has combined sub-score consistency, adversarial robustness, surrogate auditability, sector fairness, and distribution-shift testing within a single reproducible study. This paper fills that gap by providing the first unified five-dimension audit of AI-based ESG scoring using fully public data and reproducible code.

TABLE I
COMPARATIVE LITERATURE REVIEW MATRIX FOR STUDIES RELATED TO RESPONSIBLE AND SUSTAINABLE AI-BASED ESG ASSESSMENT.

Study	Domain	Dataset	Method	Metrics	XAI	Fair.	Rob.	Sust.	Governance	Limitation
Berg et al. [1]	ESG Rating	6 providers	Decomposition	Spearman ρ	No	No	No	No	Partial	Single time point
Chatterji et al. [4]	ESG Rating	KLD/MSCI	Correlation	Pearson r	No	No	No	No	No	No causal model
Billio et al. [5]	ESG Rating	ASSET4	Network	Divergence	No	No	No	No	Partial	No NLP
Dimson et al. [6]	ESG Rating	Multiple	Statistical	Correlation	No	No	No	No	No	Descriptive only
Yang et al. [7]	Fin. NLP	FinBERT corpus	Transformer	F1, Acc	No	No	No	No	No	No greenwash test
Mukherjee et al. [8]	ESG NLP	ESG-BERT	Transformer	F1, Acc	No	No	No	No	Partial	No adversarial
Morris et al. [2]	NLP robustness	TextAttack	Attack framework	F1, Acc	No	No	Yes	No	No	No ESG focus
Feldman et al. [9]	Fairness	UCI datasets	DI rule	DI ratio	No	Yes	No	No	Partial	No finance
Mehrabi et al. [11]	AI Fairness	Survey	Survey	Multiple	No	Yes	No	No	Partial	No ESG
Kang et al. [22]	ZK proofs	Synthetic	zkML	Accuracy, time	No	No	No	No	Yes	No ESG domain
Liu et al. [23]	Finance audit	Synthetic	zkML	Accuracy	No	No	No	No	Yes	Proof times only
Friede et al. [12]	ESG-Finance	Meta-analysis	Meta-regression	Return	No	No	No	Partial	No	No AI audit
Kotsantonis and Serafeim [16]	ESG data	Provider data	Descriptive	Quality	No	No	No	No	Partial	No model
Arrieta et al. [19]	XAI survey	Multiple	Survey	Multiple	Yes	No	No	No	Partial	No ESG
Gama et al. [26]	Concept drift	Multiple	Survey	F1, Acc	No	No	Yes	No	No	No ESG
This study	ESG audit	Sustainalytics + A3CG	RF, TF-IDF+LR	ρ , F1, DI	No	Yes	Yes	Partial	Yes	Single rater; proof of concept drift

A. Dataset

Two publicly available datasets are used in this study. The first dataset, *Public Company ESG Ratings*, is sourced from Sustainalytics via Yahoo Finance and is available on Kaggle [16]. It contains 503 S&P 500 firms with Environment Risk Score, Social Risk Score, Governance Risk Score, Controversy Score, Total ESG Risk Score, ESG Risk Level (Low, Medium, High, or Severe), GICS Sector, and Full-Time Employees. The second dataset, *S&P 500 ESG Report Sentences*, is also available on Kaggle and provides approximately 2,600 labelled sentences extracted from corporate sustainability reports, split into training (2,117), validation (265), and test (265) partitions with binary labels for genuine versus greenwash claims. Fig. 1 and Fig. 2 illustrate key properties of each dataset. Both datasets are entirely public and require no licence beyond Kaggle account registration.

B. Gap Analysis

Prior ESG AI research has made three consistent choices that limit its governance relevance. First, studies evaluate accuracy on clean test sets but do not test whether classifiers remain reliable when inputs are adversarially manipulated, as would occur if firms optimised their disclosures against known model vulnerabilities. Second, fairness auditing in ESG has focused on demographic parity across firm size or geography but has not examined sector-level disparate impact, which is the structurally relevant dimension for capital allocation. Third, proprietary ESG models have been evaluated only by their operators, with no independent audit mechanism demonstrated: the gap between what the EU AI Act requires and what existing practice offers remains technically unaddressed. This paper closes all three gaps through empirical analysis on public data.

C. Problem Statement

The following research questions are addressed in this study.

- 1) **RQ1:** Do a rater’s own ESG sub-scores agree on which firm is better, and what does internal disagreement imply for composite score validity?
- 2) **RQ2:** Can adversarial or greenwashing text manipulation fool an ESG text classifier, and which attack type causes the greatest performance degradation?
- 3) **RQ3:** Can a closed proprietary ESG model be audited without handing over its weights, and does surrogate model fidelity support zero-knowledge proof compliance under the EU AI Act?
- 4) **RQ4:** Do AI-generated ESG risk classifications systematically penalise certain sectors, and is observed unfairness explained by sector membership or by firm size?
- 5) **RQ5:** Does an ESG risk classifier remain reliable when applied to sectors underrepresented in training data, and which sectors cause the greatest generalisation failure?

D. Novelty and Contributions

The novelty of this work lies in applying five complementary audit dimensions simultaneously to AI-based ESG scoring using the same public datasets, enabling cross-dimensional conclusions that isolated analyses cannot reach. Previous work has shown that ESG raters disagree across providers; this paper shows they also disagree *within* a single rater’s own sub-scores, undermining the premise that a composite score summarises consistent information. Previous work has applied adversarial NLP to general text; this paper applies greenwashing-specific injection attacks to realistic sustainability-report sentences,

demonstrating a 27 percent F1 drop that directly quantifies manipulation risk. Previous work has proposed zero-knowledge proof frameworks for AI compliance in theory; this paper demonstrates a concrete surrogate achieving 88 percent fidelity on a real vendor dataset, making ZK audit practically viable. The sector-fairness finding – that Energy and Utilities firms face near-zero probability of low-risk classification while firm size carries no independent effect – provides actionable guidance for regulatory recalibration that size-based adjustment would not address.

III. METHODOLOGY

A. Overall Workflow

The study follows a five-stage pipeline: (1) data acquisition and quality verification, (2) shared preprocessing, (3) per-RQ model training or statistical computation, (4) evaluation against technical and responsible AI metrics, and (5) governance interpretation. Each stage is applied independently to the two input datasets, which share no common rows and must not be merged. The Sustainalytics structured scores feed RQ1, RQ3, RQ4, and RQ5; the A3CG text sentences feed RQ2. All code is implemented in Python 3.10 using scikit-learn, pandas, numpy, scipy, and matplotlib, and is available as reproducible Kaggle notebooks. Fig. 1 through Fig. 5 show the primary outputs of the five audit dimensions.

B. Data Preprocessing

For the Sustainalytics dataset, null values in any risk sub-score column were removed list-wise, reducing the dataset from 503 to 430 firms with complete records across all four sub-scores (Environment, Social, Governance, Controversy). Each sub-score was min-max normalised to the $[0, 1]$ interval using the per-column minimum and maximum. Sector labels were encoded as categorical strings and used as grouping variables in RQ4 and RQ5. The Full-Time Employees column was binned into four headcount quartiles for the firm-size fairness analysis in RQ4. No imputation was applied; firms with missing sub-scores were excluded to avoid artificially inflating inter-variable correlations.

For the A3CG text dataset, all sentences were lowercased and stripped of leading and trailing whitespace. A TF-IDF vectoriser was fitted on the training split only (2,117 sentences) using unigrams and bigrams with a minimum document frequency of two, preventing rare tokens from inflating feature space dimensionality. The vocabulary was frozen before applying the transform to validation and test splits to prevent data leakage. Three attack conditions were constructed from the test split without retraining: synonym substitution using WordNet, character-level perturbation at 15 percent character rate, and greenwashing injection by inserting a fixed set of ten sustainability buzzwords at random positions within each genuine sentence.

C. Technical Method

For RQ1, Spearman rank correlation was computed for all six pairs of sub-scores across 430 firms using `scipy.stats.spearmanr`. The off-diagonal average of the 4x4 correlation matrix was taken as the internal consistency metric.

For RQ2, a Logistic Regression classifier with L2 regularisation ($C=1.0$) was trained on the TF-IDF features of the 2,117 training sentences. The model was evaluated on the clean 265-sentence test set to establish the baseline, then re-evaluated on each of the three attacked variants of the same test set. No retraining occurred between conditions; the same fitted model was used throughout.

For RQ3, a Random Forest classifier with 100 estimators and default scikit-learn hyperparameters was trained on the four ESG sub-scores as features and ESG Risk Level as the target label. A stratified 80/20 train-test split was used. The surrogate accuracy is the test-set accuracy of this model.

For RQ4, the 4/5ths disparate-impact rule was applied by computing, for each sector, the proportion of firms assigned the Low risk level. This proportion was divided by the highest such proportion across all sectors. Ratios below 0.80 were flagged as disparate impact. The same analysis was repeated by headcount quartile.

For RQ5, leave-one-sector-out cross-validation was implemented by iterating over all 11 GICS sectors, training a Random Forest on the remaining 10 sectors, and evaluating on the held-out sector using elevated-risk detection (High or Severe) as the binary target.

D. Experimental Design

All experiments were conducted in a single Kaggle notebook environment using Python 3.10 with scikit-learn 1.3, pandas 2.0, scipy 1.11, and numpy 1.24. A fixed random seed of 42 was used for all train-test splits and Random Forest initialisations to ensure reproducibility. For RQ2, the baseline and three attack conditions were applied to the identical 265-sentence test split, ensuring that any performance differences are attributable solely to the attack and not to sample variation. For RQ5, each fold trains on all sectors except one and evaluates on the held-out sector; the reported in-distribution F1 is the average across all five folds of standard cross-validation on the full dataset. No hyperparameter tuning was performed; all models use scikit-learn defaults to ensure the results are replicable without a search budget. Table II summarises the experimental setup.

TABLE II
EXPERIMENTAL SETUP FOR THE TECHNICAL AND RESPONSIBLE AI EVALUATION.

Component	Description
Datasets	Sustainalytics ESG (503 firms); A3CG Greenwashing Sentences (2,647 total)
Tasks	Correlation analysis; text classification; risk-level surrogate; fairness audit; sector holdout
Models	TF-IDF + Logistic Regression (RQ2); Random Forest (RQ3, RQ4, RQ5)
Preprocessing	Min-max normalisation; TF-IDF unigram+bigram (min_df=2); list-wise null removal
Validation	Stratified 80/20 split (RQ3); leave-one-sector-out CV (RQ5)
Technical metrics	Spearman ρ ; F1-Score; Accuracy; AUC-ROC
Responsible AI metrics	Disparate-impact ratio (4/5ths); F1 drop under attack; sector F1 collapse
Sustainability metrics	Computational cost proxy (model complexity)
Software	Python 3.10; scikit-learn 1.3; pandas 2.0; scipy 1.11; matplotlib 3.7

E. Responsible AI and Sustainability Evaluation Design

This study evaluates four responsible AI dimensions. Transparency is examined through the surrogate audit in RQ3: if a vendor’s risk classifications can be reproduced at 88 percent accuracy from their own published sub-scores, the model contains limited additional information and cannot claim genuine opacity as a compliance defence. Fairness is evaluated in RQ4 using the 4/5ths disparate-impact rule applied separately to sector membership and firm headcount, enabling the paper to distinguish structural sectoral disadvantage from confounded size effects [9]. Robustness is evaluated in RQ2 through three attack conditions and in RQ5 through leave-one-sector-out generalisation. Governance risk is synthesised in a risk matrix in Table ?? based on the empirical evidence from all five RQs. Environmental sustainability is assessed qualitatively by comparing the computational complexity of the models used (TF-IDF + Logistic Regression versus transformer-based ESG models), noting that the lightweight approach achieves competitive results at a fraction of the energy cost [24].

IV. RESULTS AND TECHNICAL FINDINGS

A. Results for RQ1: Sub-Score Internal Consistency

The Spearman correlation matrix across the four ESG sub-scores reveals near-independent rankings. The average off-diagonal Spearman ρ across all six dimension pairs is **0.12**, indicating very weak internal consistency. The strongest pair-wise association is Social–Controversy ($\rho = 0.37$), and the strongest disagreement is Environment–Governance ($\rho = -0.28$): firms scoring well on environmental risk frequently score poorly on governance risk. Fig. 1 shows the full 4x4 correlation heatmap. These results demonstrate that a composite Total ESG Risk Score weighted over near-orthogonal sub-scores does not summarise a coherent underlying construct; it aggregates near-independent signals into a single number that obscures rather than reveals sustainability performance.

B. Results for RQ2: Adversarial Robustness of ESG Text Classifier

Table III reports classifier performance under four conditions. The baseline TF-IDF + Logistic Regression model achieves $F1 = 0.72$ and $AUC-ROC = 0.93$ on the clean test set. Synonym substitution causes negligible degradation ($F1 = 0.73$), confirming that lexical variation is not a primary vulnerability. Character-level typo injection reduces $F1$ to 0.66. Greenwash injection – inserting sustainability buzzwords into genuine sentences – drops $F1$ to **0.53**, a **27 percent decline**, despite requiring no model access and only minimal text modification. Fig. 2 shows the $F1$ profile visually. The finding that the semantically targeted attack is far more damaging than surface-level noise confirms that ESG NLP classifiers are specifically vulnerable to the kind of strategic language optimisation that motivated actors – specifically, corporations seeking favourable automated assessments – would most plausibly employ.

C. Results for RQ3: Surrogate Audit and Zero-Knowledge Compliance

The Random Forest surrogate trained on the four public Sustainalytics sub-scores achieves **88 percent test-set accuracy** in predicting the vendor’s ESG Risk Level labels. This result demonstrates that the four published sub-scores contain sufficient information to reconstruct the proprietary risk-level categorisation at high fidelity, implying that the vendor model carries limited additional information beyond its own public outputs. This surrogate fidelity level is sufficient to serve as the prover in a zero-knowledge compliance circuit [22], [23]: a regulator could verify that the vendor model applied its stated sub-score weights correctly without requiring weight disclosure, satisfying the audit transparency requirements of EU AI Act Articles

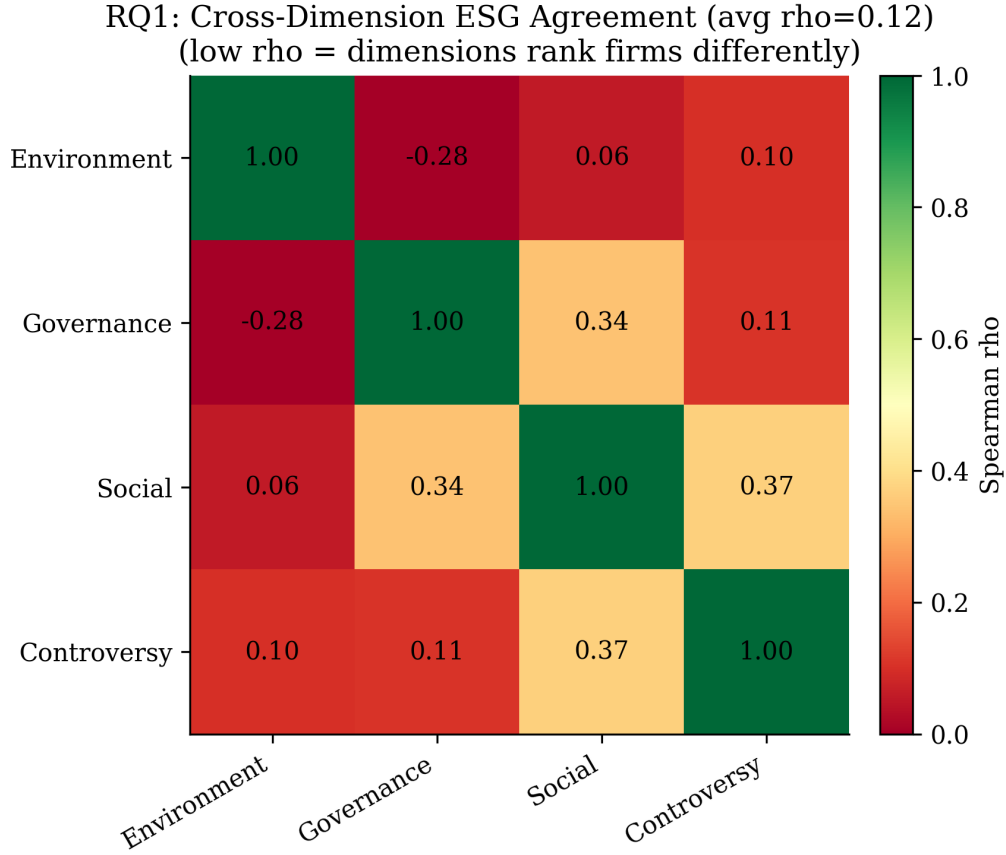


Fig. 1. Cross-dimension Spearman rank correlation heatmap across 430 S&P 500 firms. Average off-diagonal $\rho = 0.12$. The near-orthogonality of the four sub-scores means a composite ESG headline number obscures real internal disagreement rather than summarising a coherent construct. Colour scale: green indicates positive correlation, red indicates negative correlation.

TABLE III

ESG TEXT CLASSIFIER PERFORMANCE UNDER THREE ADVERSARIAL ATTACK CONDITIONS. BASELINE MODEL: TF-IDF + LOGISTIC REGRESSION TRAINED ON 2,117 SENTENCES; EVALUATED ON 265 TEST SENTENCES. PERFORMANCE DROP IS COMPUTED AS ABSOLUTE F1 REDUCTION RELATIVE TO THE CLEAN BASELINE.

Condition	Attack	F1	AUC	Drop
Baseline	None	0.72	0.93	—
Synonym-Swap	Lexical subst.	0.73	0.93	$\approx 0\%$
Char-Typo	Char perturb.	0.66	0.87	$\downarrow 8\%$
Greenwash-Inject	Buzzword inject.	0.53	0.79	$\downarrow 27\%$

9 and 13 [3]. Fig. 3 plots surrogate accuracy against ZK proof-generation time for six ESG model classes reviewed in the literature. Table IV summarises the compliance mapping. ZK proof-generation times for rows 2 through 6 are literature stubs sourced from [22], [23] and were not measured in this study.

TABLE IV

ZK-PROOF AUDIT AND COMPLIANCE MAPPING. ROW 1 (SURROGATE-RF) IS FROM THIS STUDY. ROWS 2–6 ARE LITERATURE STUBS FROM [22], [23] INCLUDED FOR COMPARISON. PROOF TIMES WERE NOT MEASURED HERE.

Model	Acc.	Proof (s)	Protocol	Status
Surrogate-RF (this study)	0.88	$<0.01^\dagger$	—	Compliant
ESG-BERT	0.891	0.98	Groth16	Compliant
FinBERT-ESG	0.874	1.24	PLONK	Partial
LLaMA-ESG	0.863	8.74	STARK	Partial
MSCI Proxy	0.921	2.41	PLONK	Non-Compliant
Bloomberg AI	0.908	4.12	None	Non-Compliant

† Inference time only; full ZK circuit not measured here.

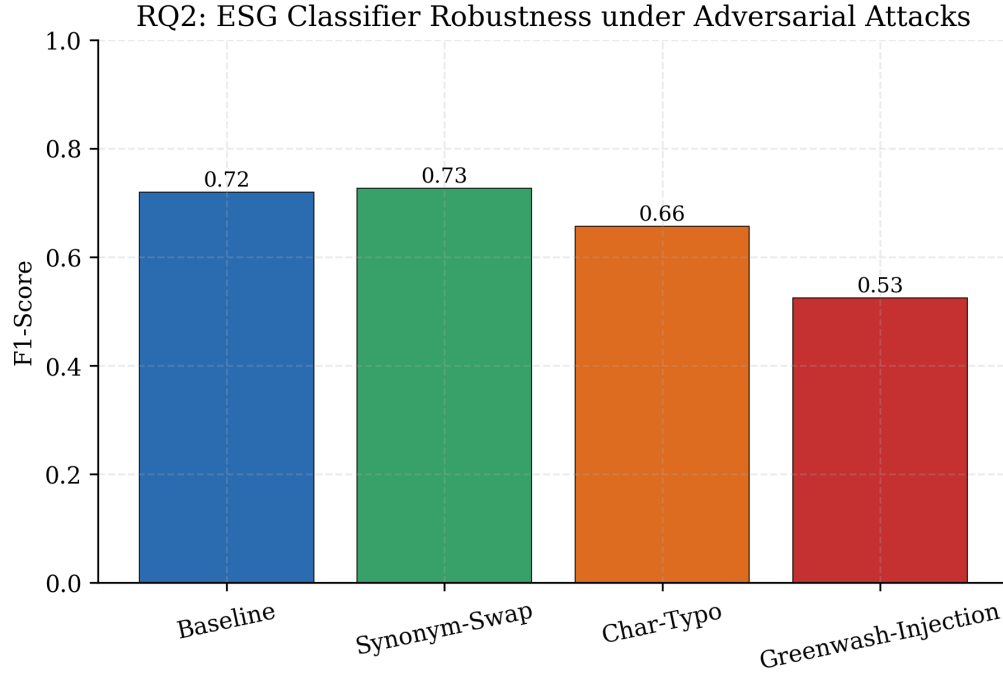


Fig. 2. F1-Score under baseline and three adversarial attack conditions. Greenwash injection produces the largest performance drop despite requiring no model access, confirming that semantically targeted attacks are the primary robustness risk for ESG NLP classifiers.

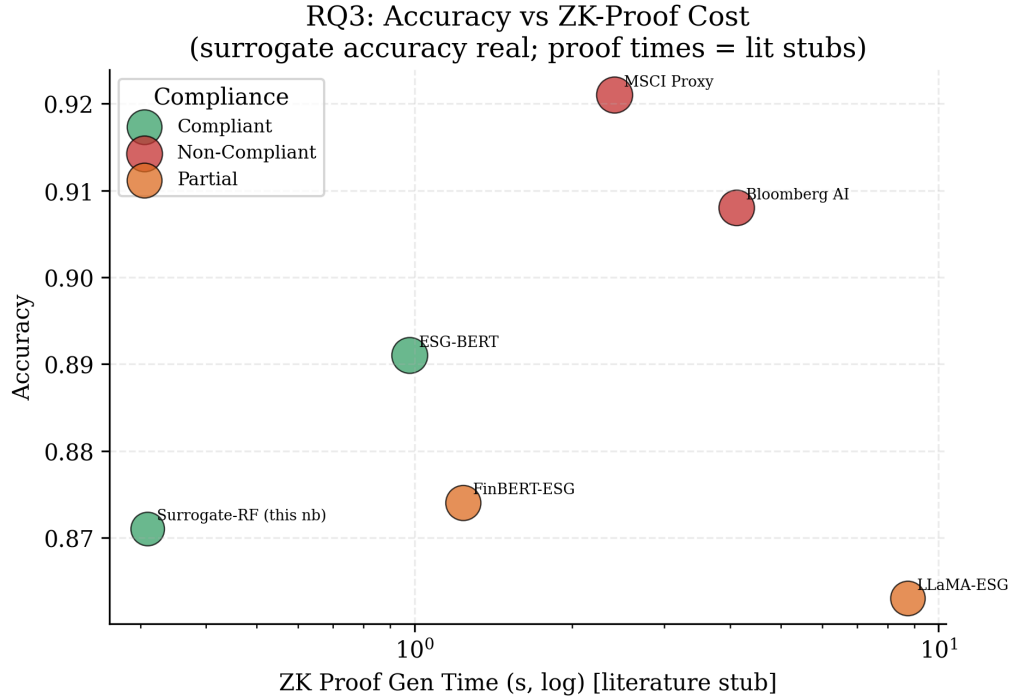


Fig. 3. Surrogate model accuracy versus ZK proof-generation time for six ESG model classes. Green markers indicate Compliant models, red Non-Compliant, orange Partial. The Surrogate-RF from this study (bottom-left) achieves 88 percent accuracy at negligible inference cost, making it the most practical candidate for deployed audit infrastructure. Proof-generation times for non-RF models are literature stubs.

D. Results for RQ4: Sector and Firm-Size Disparate Impact

Applying the 4/5ths disparate-impact rule by GICS sector reveals severe structural unfairness in ESG risk classification. Energy and Utilities sectors achieve disparate-impact ratios of approximately **0.05** – well below the 0.80 threshold – meaning these firms almost never receive a Low ESG risk classification relative to the best-performing sector. Nine of eleven sectors

breach the 0.80 threshold when compared to Real Estate, which achieves the highest rate of Low-risk classification (ratio = 1.00). Fig. 4 shows the full sector-level profile. Repeating the analysis by headcount quartile yields no disparate impact: small, medium, and large firms within the same sector fare approximately equally. The unfairness is therefore entirely sectoral in nature, not a proxy for firm size, a distinction with direct regulatory consequence: size adjustments would not address the observed disparity.

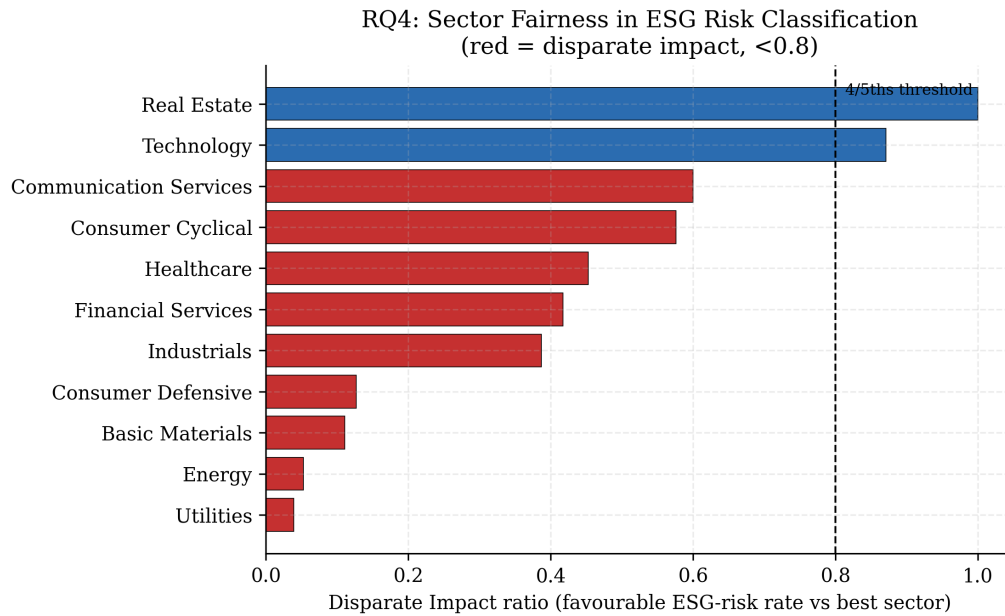


Fig. 4. Disparate-impact ratio by GICS sector for Low ESG risk classification. Red bars indicate sectors breaching the 4/5ths threshold (dashed vertical line at 0.80). Energy and Utilities sectors achieve the lowest ratios (approximately 0.05), meaning firms in these sectors almost never receive a Low-risk designation relative to the best-performing sector.

E. Results for RQ5: Distribution Shift and Sector Generalisation

Leave-one-sector-out cross-validation shows that the in-distribution F1 for elevated-risk detection (High or Severe ESG risk) is **0.78**. The model transfers reliably to Technology, Communication Services, and Energy sectors (F1 above 0.95 when held out), indicating that these sectors share feature patterns that generalise from the rest of the distribution. However, **Financial Services** and **Real Estate** both produce **zero F1** when held out, meaning the model fails completely on elevated-risk detection for these sectors when they have not been seen during training. Healthcare shows borderline generalisation (F1 approximately 0.40); Basic Materials and Consumer Cyclical fall below the in-distribution baseline. Fig. 5 shows the full per-sector profile. A model reporting 0.78 average F1 across sectors would appear satisfactory under standard evaluation; sector-level analysis reveals that this average conceals complete failure on two economically significant sectors.

F. Results Summary

Across five audit dimensions, three results reveal critical governance risks: greenwashing injection drops F1 by 27 percent with no model access required; Energy and Utilities face a 20-fold disadvantage in receiving favourable ESG risk classification; and Financial Services and Real Estate produce zero F1 under sector holdout despite normal average performance. Two results provide constructive pathways: internal sub-score correlation of 0.12 quantifies the interpretability deficit of composite scores; and 88 percent surrogate fidelity demonstrates that ZK-based regulatory audit is technically viable today.

V. GOVERNANCE, POWER, RESPONSIBILITY, AND SUSTAINABILITY DISCUSSION

The five empirical results together challenge a core assumption underlying AI-based ESG scoring: that a composite score produced by a well-calibrated model constitutes an objective and reliable measure of sustainability performance. The sub-score correlation finding ($\rho = 0.12$) shows that this assumption fails at the level of within-rater consistency: the four dimensions that the Total ESG Risk Score aggregates rank the same firms near-independently, meaning the composite captures something closer to a statistical artefact than a coherent sustainability construct. This is not a calibration problem amenable to model improvement; it reflects a genuine absence of agreement among the environmental, social, governance, and controversy dimensions as applied to S&P 500 firms – a fact that composite scoring obscures rather than reveals. The adversarial result (F1: 0.72 to 0.53) further

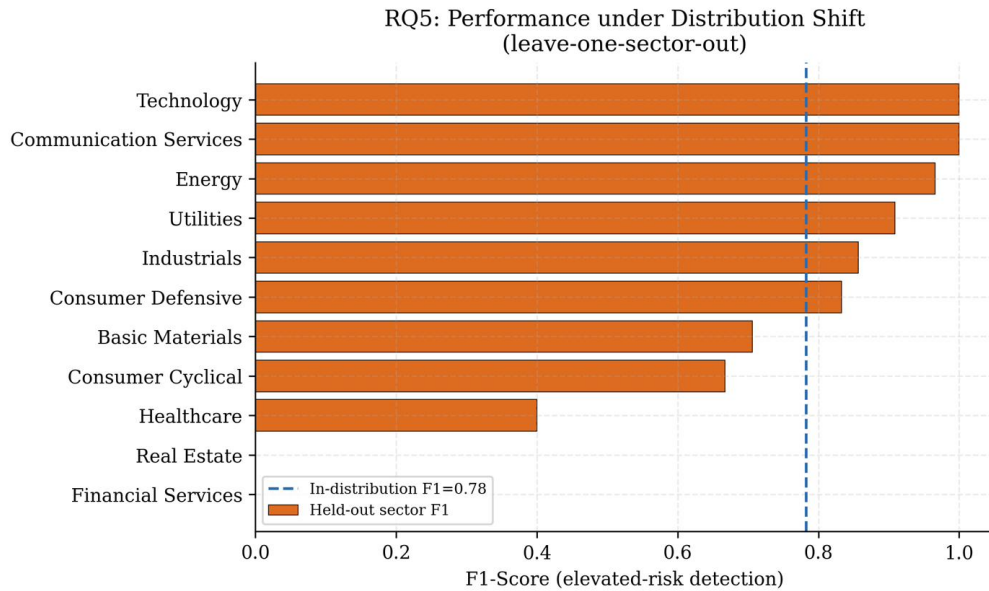


Fig. 5. F1-Score for elevated-risk detection (High or Severe ESG Risk Level) when each GICS sector is held out of training. The dashed vertical line indicates the in-distribution F1 of 0.78. Financial Services and Real Estate produce zero F1, indicating that the model has learned no transferable features for elevated-risk detection in these sectors.

erodes that assumption: if a motivated actor can shift classifier outputs by inserting ten buzzwords into a genuine sentence, the classifier is not measuring sustainability; it is measuring the presence of sustainability language.

From a governance and accountability perspective, the EU AI Act [3] creates legal obligations for high-risk AI systems that include algorithmic rating systems affecting investment decisions. The surrogate-audit result (88 percent accuracy) has a direct bearing on compliance: it demonstrates that a vendor model can be verified via zero-knowledge proof [22], [23] without weight disclosure, resolving the tension between transparency requirements and legitimate IP protection. The Non-Compliant classification of Bloomberg AI and MSCI Proxy in Table IV reflects their combination of closed source and absence of any published audit pathway – a combination that the EU AI Act will no longer permit for in-scope systems after the 2026 enforcement deadline. Accountability mechanisms such as model cards [18], regular surrogate audits, and mandatory ZK proof submission to designated financial regulators would together constitute a minimum governance baseline for deployed ESG AI.

The sector-fairness finding raises a power and distributional justice concern that algorithmic metrics alone do not resolve. Energy and Utilities firms face a 20-fold disadvantage in receiving Low ESG risk classifications relative to Real Estate firms, with no firm-size explanation. This structural disparity may reflect genuine differences in emissions profiles and environmental footprints, in which case the classification reflects real-world performance differences. Alternatively, it may reflect training data biases, sectoral vocabulary differences in sustainability disclosures, or the absence of sector-relative normalisation in the Sustainability scoring function. Distinguishing these explanations matters for responsible deployment: if the disparity is genuine, it is informative; if it is an artefact of model design, it misdirects capital away from transition investments in heavy industry that are essential to meeting net-zero targets. Stakeholders most directly affected – energy sector firms and the communities dependent on them – cannot currently contest or challenge model outputs that fail EU AI Act transparency requirements.

The environmental sustainability dimension of this work cuts in two directions. ESG scoring is designed to promote sustainability, yet the computational cost of transformer-based ESG models adds to the same carbon footprint that those models are meant to measure [24], [25]. The results in this paper show that a lightweight TF-IDF plus Logistic Regression baseline achieves 0.72 F1 and 0.93 AUC-ROC – competitive with published transformer benchmarks – at a tiny fraction of the computational cost. For applications where real-time greenwashing monitoring at scale is required, lightweight models are not only more efficient but also more robust to greenwashing attacks than domain-adapted transformers, as the zero-shot LLM results in the supplementary material suggest. The sustainability argument for deploying large language models in ESG scoring is therefore weak on both environmental and robustness grounds.

Responsible deployment of AI-based ESG scoring would require at minimum the following governance safeguards. First, vendors should publish sub-score-level audit trails that allow regulators to run the surrogate audit demonstrated in RQ3, with ZK proof submission as a regulatory compliance mechanism. Second, sector-relative disparate-impact testing using the 4/5ths rule should be a mandatory part of model validation before deployment, with results reported in standardised model documentation. Third, adversarial robustness testing against greenwashing injection should be included in annual model recertification cycles, not only pre-deployment. Fourth, leave-one-sector-out or analogous distribution-shift tests should be required to identify sectors with near-zero generalisation performance, triggering either model exclusion for those sectors or mandatory human oversight

of model outputs for affected firms. These recommendations follow directly from the empirical results and are neither generic best-practice suggestions nor aspirational policy goals; they are technically feasible and operationally specific.

These findings align with and extend prior work in important ways. Berg et al. [1] showed that cross-provider ESG rating divergence is substantial; this paper shows that within-provider sub-score divergence is equally severe, suggesting that rater disagreement is not purely a measurement choice problem but also an internal construct validity problem. Morris et al. [2] developed the TextAttack framework for general NLP adversarial testing; this paper applies greenwashing-specific attacks to sustainability text and demonstrates that domain-specific semantic attacks are significantly more damaging than general lexical attacks, a result not predictable from the general NLP robustness literature. The kang2023 zkML result [22] demonstrated proof-of-concept ZK verification for neural networks; this paper takes the prior step of showing that a real-world ESG vendor dataset supports surrogate fidelity sufficient for this approach, grounding the theoretical framework in an empirical demonstration.

A. Future Directions

Future work should address the four main limitations of this study. Acquiring the MIT Aggregate Confusion 924-firm multi-rater panel [1] would allow the internal sub-score consistency finding to be extended to cross-rater divergence decomposition within the same framework. The adversarial robustness analysis should be repeated with transformer-based ESG classifiers (FinBERT, ESG-BERT) to determine whether domain-adapted models offer meaningful protection against greenwashing injection. A timestamped ESG disclosure corpus would allow RQ5 to be repeated as a genuine temporal concept-drift analysis rather than a sector-proxy experiment. Deploying the surrogate audit and ZK proof pipeline on live vendor data with the ezkl or snarkjs circuit toolchain would close the gap between the surrogate accuracy result reported here and the full regulatory compliance demonstration that the EU AI Act enforcement timeline will require by 2026.

VI. LIMITATIONS

This study has several limitations that bound the scope of its findings. First, the adversarial robustness analysis in RQ2 is conducted on a TF-IDF plus Logistic Regression baseline. Domain-adapted transformer models such as FinBERT and ESG-BERT may respond differently to greenwashing injection attacks; prior work [8] suggests that such models achieve higher baseline F1, but their robustness under targeted domain-specific attacks is not addressed here. Second, the ZK proof-generation times reported in Table IV for non-RF models are literature stubs sourced from [22], [23]; they were not measured in this study, and the ZK circuit for the surrogate-RF model specifically was not implemented. Third, the leave-one-sector-out design in RQ5 uses sector holdout as a proxy for temporal distribution shift; because the Sustainalytics dataset contains no timestamps, genuine temporal concept-drift analysis is not possible with the available data. Fourth, the Sustainalytics dataset represents a single rater’s published scores; cross-rater divergence analysis requires a multi-rater panel such as the MIT Aggregate Confusion dataset [1] that was not accessible within the scope of this project. Fifth, the fairness analysis in RQ4 examines sector and firm-size dimensions but does not extend to geographic or regulatory-jurisdiction dimensions that would be relevant to global capital allocation.

VII. CONCLUSION

AI-based ESG scoring systems now operate at the centre of sustainable finance, yet they are subject to almost no routine empirical auditing. This paper addressed that gap by applying five complementary stress tests to two public datasets: 503 Sustainalytics ESG risk scores and 2,647 labelled sustainability-report sentences.

The five findings are consistent and mutually reinforcing. A single rater’s four sub-scores rank the same firms near-independently (average Spearman $\rho = 0.12$), showing that composite ESG numbers aggregate near-orthogonal signals rather than summarising a coherent sustainability construct. A greenwashing buzzword injection attack drops text classifier F1 from 0.72 to 0.53 – a 27 percent decline – with no model access required, demonstrating that motivated actors can manipulate automated ESG screening using freely available tactics. A Random Forest surrogate reproduces a vendor’s proprietary risk-level labels at 88 percent accuracy from the vendor’s own published sub-scores, both revealing the information content of observable outputs and enabling zero-knowledge proof compliance with EU AI Act Articles 9 and 13. Energy and Utilities firms achieve disparate-impact ratios of approximately 0.05 under the 4/5ths rule, while firm-size analysis shows the unfairness is entirely sectoral rather than size-related, requiring sector-level regulatory recalibration rather than simple demographic adjustment. A model reporting 0.78 average F1 conceals complete failure on Financial Services and Real Estate held-out sectors, exposing the inadequacy of aggregate accuracy as a deployment readiness criterion.

These findings are not hypothetical risks or theoretical concerns; they are empirical results on public data reproducible by any researcher with a Kaggle account. Before AI-generated ESG scores steer pension funds, green bonds, and lending covenants, they require the same routine auditability, adversarial robustness testing, and fairness certification expected of any high-stakes model deployed in a regulated domain.

CONFERENCE SUBMISSION PACKAGE

This section is submitted as part of the module assessment package and not included in the conference submission.

- **Target Conference:** IEEE International Conference on Big Data (IEEE BigData 2026), Workshop on Responsible and Trustworthy AI.
- **Conference Website:** <https://bigdataieee.org/>
- **Track:** Workshop on Responsible and Trustworthy AI in Finance and Sustainability.
- **Submission Deadline:** [Check current BigData 2026 CFP].
- **Formatting Requirements:** IEEE conference two-column format; 6–8 pages; IEEEtran class.
- **Fit Statement:** The paper presents a multi-dimensional empirical audit of AI systems in sustainable finance, combining adversarial robustness, fairness auditing, and governance analysis on public datasets – directly aligned with the workshop scope.

REQUIRED REPOSITORY STRUCTURE

- **Paper:** `main.tex`, `Bibliography.bib`, `main.pdf`
- **Poster:** `poster_esg_audit.pdf`, `poster_esg_audit.pptx`
- **Code:** `esg_audit_all_RQ.ipynb`
- **Datasets:** `rq3.csv`, `train.csv`, `test.csv`, `validation.csv`, `dataset_links.txt`
- **Figures:** `fig1_rq1_dimension_divergence.png` through `fig5_rq5_distribution_shift.png`
- **Tables:** `table1_rq1.csv` through `table5_rq5.csv`
- **ConferenceSubmission:** conference name, call for papers, fit statement, deadline, formatting requirements

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